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United States Patent	12404780
Kind Code	B2
Date of Patent	September 02, 2025
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Containment ring for gas turbine engine

Abstract

A containment ring for a turbine casing assembly, including: a main body portion, the main body portion having an outer periphery and an inner periphery and a forward end and an aft end each extending between the outer periphery and the inner periphery, the main body portion also including an outer containment zone and an inner containment zone, the inner containment zone is integrally formed with the outer containment zone in order to provide a single unitary structure and the inner containment zone being radially inward from the outer containment zone, the inner containment zone extending radially outward from the inner periphery and having a plurality of ribs extending between the forward end and the aft end.

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Appl. No.: 18/425993

Filed: January 29, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250243778 A1	Jul. 31, 2025

Publication Classification

Int. Cl.: F01D21/04 (20060101); F01D25/24 (20060101); F02C7/00 (20060101)

U.S. Cl.:

Field of Classification Search**CPC:** F01D (21/045)

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Background/Summary

BACKGROUND

(1) This disclosure relates to structures for use in gas turbine engines, and more particularly to a containment ring for use in a gas turbine engine.

(2) Gas turbine engines have a turbine casing assembly. The turbine casing assembly may include a containment ring. It is desirable to reduce the weight of the containment ring while maintaining its performance.

BRIEF DESCRIPTION

(3) Disclosed is a containment ring for a turbine casing assembly, including: a main body portion, the main body portion having an outer periphery and an inner periphery and a forward end and an aft end each extending between the outer periphery and the inner periphery, the main body portion also including an outer containment zone and an inner containment zone, the inner containment zone is integrally formed with the outer containment zone in order to provide a single unitary structure and the inner containment zone being radially inward from the outer containment zone, the inner containment zone extending radially outward from the inner periphery and having a plurality of ribs extending between the forward end and the aft end.

(4) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the plurality of ribs are equally spaced from each other.

(5) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the plurality of ribs are angularly offset with respect to an inner axis of the containment ring.

(6) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the plurality of ribs are angularly offset at angle with respect to an inner axis of the containment ring, the angle being within a range of 20 degrees to 70 degrees with respect to an inner axis of the containment ring.

(7) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the plurality of ribs are angularly offset at a 45 degree angle with respect to an inner axis of the containment ring.

(8) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the containment ring further including a flange portion secured to the main body portion.

(9) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the flange portion is secured to the main body portion via a plurality of tabs.

(10) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the main body portion is circular in shape.

(11) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the plurality of ribs are each separated by an opening.

(12) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the opening does not extend to the outer periphery or the forward end or the aft end.

(13) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the opening extends from the inner periphery and has a scalloped shape.

(14) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the opening has an inner surface that is furthest from the inner periphery at a central portion of the opening and a depth of the opening from the inner periphery gradually decreases as it extends from the central portion towards the forward end or the aft end.

- (15) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the opening has a curved bottom surface that extends from the central portion towards the forward end and aft end of the main body portion.
- (16) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, an axial length of the inner periphery from the forward end to the aft end is greater than an axial length of the outer periphery from the forward end to the aft end.
- (17) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the main body portion has a trapezoidal shape in its cross section.
- (18) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the forward end and the aft end are angled surfaces that extend towards each other from the inner periphery.
- (19) Also disclosed is a turbine casing assembly, including: an outer structural case; and a containment ring, including: a main body portion, the main body portion having an outer periphery and an inner periphery and a forward end and an aft end each extending between the outer periphery and the inner periphery, the main body portion also including an outer containment zone and an inner containment zone, the inner containment zone is integrally formed with the outer containment zone in order to provide a single unitary structure and the inner containment zone being radially inward from the outer containment zone, the inner containment zone extending radially outward from the inner periphery and having a plurality of ribs extending between the forward end and the aft end.
- (20) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the inner containment zone is integrally formed with the outer containment zone in order to provide a single unitary structure and the plurality of ribs are angularly offset with respect to an inner axis of the containment ring.
- (21) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the plurality of ribs are angularly offset at angle with respect to an inner axis of the containment ring, the angle being within a range of 20 degrees to 70 degrees with respect to an inner axis of the containment ring and the plurality of ribs are each separated by an opening.
- (22) Also disclosed is a gas turbine engine, including: a compressor section; a combustor; a turbine section; and a turbine casing assembly, including: an outer structural case; and a containment ring, including: a main body portion, the main body portion having an outer periphery and an inner periphery and a forward end and an aft end each extending between the outer periphery and the inner periphery, the main body portion also including an outer containment zone and an inner containment zone, the inner containment zone is integrally formed with the outer containment zone in order to provide a single unitary structure and the inner containment zone is radially inward from the outer containment zone, the inner containment zone extending radially outward from the inner periphery and having a plurality of ribs extending between the forward end and the aft end.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:
- (2) FIG. 1 is a schematic cross-sectional view of a gas turbine engine;
- (3) FIG. 2 is a perspective view of a containment ring and a rotor in accordance with the present disclosure;
- (4) FIG. 3 is a perspective view of a containment ring in accordance with the present disclosure;
- (5) FIG. 4 is a perspective cross-sectional view of a containment ring in accordance with the

present disclosure;

(6) FIG. 5 is an enlarged perspective cross-sectional view of a portion of the containment ring in accordance with the present disclosure;

(7) FIG. 6 is cross-sectional view of the containment ring in accordance with the present disclosure;

(8) FIG. 7 is another cross-sectional view of the containment ring in accordance with the present disclosure; and

(9) FIG. 8 is an illustration of containment of a rotor by the containment ring in accordance with the present disclosure.

DETAILED DESCRIPTION

(10) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the FIGS.

(11) FIG. 1 illustrates a gas turbine engine **10**, generally comprising in serial flow communication a compressor section **14** for pressurizing the air, a combustor section **16** in which the compressed air is mixed with fuel and ignited for generating an annular stream of hot combustion gases, and a turbine section **18** for extracting energy from the combustion gases. Some of the rotatable components of the gas turbine engine **10** rotate about a longitudinal center axis **20** of the gas turbine engine **10**.

(12) The gas turbine engine **10** has a “cold” section **22** and a “hot” section **24**. The cold section **22** includes those components of the gas turbine engine **10** which are upstream (relative to the direction gases flow through the gas turbine engine **10**) of the combustor **16** and have thus not been exposed to the hot combustion gases. The hot section **24** includes the combustor **16** and those components of the gas turbine engine **10** which are downstream of the combustor **16**. The components of the hot section **24** are thus exposed to the hot combustion gases generated in the combustor **16**. The gases flowing through the cold section **22** have a lower temperature than the gases flowing through the hot section **24**.

(13) The hot section **24** includes the combustor **16**, the turbine section **18** and a case downstream of the turbine section **18** for conveying the exhaust gases. The turbine section **18** includes one or more rotors **26** each having a plurality of rotor blades **28** secured to a hub **29** which rotates about the center axis **20** and extract energy from the combustion gases. In one implementation the turbine blades **28** are integrally formed with the hub **29** in order to form a single component referred to as an integrally bladed rotor or IBR **26**. The hot section **24** includes stationary bodies which enclose other components of the hot section **24** and define the gas path for the hot combustion gases. These stationary bodies are sometimes referred to as casings or cases which collectively define radially-outer boundaries of the gas turbine engine **10**.

(14) Referring now to at least FIGS. 1-7, the casing of the gas turbine engine **10** includes a turbine casing assembly **30** which is part of the hot section **24**. The turbine casing assembly **30** is a group of casing components that form part of the turbine section **18** and enclose the combustion gases. The turbine casing assembly **30** may be provided as disassembled components which may then be assembled in a suitable facility. The turbine casing assembly **30** includes an outer structural case **32**, a containment ring **34**, and at least one vane ring **36**. The at least one vane ring **36** having at least one vane **38** secured thereto. In one embodiment, the least one vane ring **36** having the at least one vane **38** may be formed as a single component.

(15) In one embodiment, the outer structural case **32**, the containment ring **34**, and the least one vane ring **36** have circular portions or configurations.

(16) FIG. 2 is a perspective view of a containment ring **34** and rotor **26** in accordance with the present disclosure and FIG. 3 is a perspective view of the containment ring **34** in accordance with the present disclosure. FIG. 4 is a perspective cross-sectional view of the containment ring **34** and FIG. 5 is an enlarged perspective cross-sectional view of a portion of the containment ring **34** in accordance with the present disclosure.

(17) The containment ring **34** has a main body portion **39**. The main body portion **39** having an

outer periphery **40**, an inner periphery **42**, a forward end **44** and an aft end **46**. As used herein, the outer periphery **40** is radially outward from the inner periphery **42**. In addition, and when the containment ring is installed in the engine **10**, the outer periphery **40** is radially further from the central axis **20** than the inner periphery **42**. The containment ring **34** also has a flange portion **48**, that is configured to secure the containment ring **34** to the outer structural case **32**. The flange portion **48** being secured to the main body portion **39**. In one non-limiting embodiment, the containment rings **34** and/or the flange portion **48** may be formed from nickel based alloys or any other suitable material.

(18) In addition, the containment ring **34** also has an outer portion or outer containment zone **50** and an inner portion or inner containment zone **52**. While two zones **50** and **52** are mentioned it is understood that zones **50** and **52** are contiguous or in other words zones **50** and **52** are part of a single unitary structure comprising at least the main body portion **39**. The outer portion or outer containment zone **50** being radially outward from the inner portion or inner containment zone **52** and the outer portion or outer containment zone **50** being integrally formed with the inner portion or inner containment zone **52** such that the containment ring **34** is a single unitary structure with the outer portion or outer containment zone **50** and the inner portion or inner containment zone **52**.

(19) The inner portion or inner containment zone **52** of the containment ring **34** includes a plurality of equally spaced ribs **54** oriented at an angle offset from the axis of rotation **20**. It is also understood and in another embodiment, that the ribs **54** could also be unequally spaced from each other. In one embodiment, the ribs **54** are angularly offset at angle with respect to an inner axis of the containment ring, the angle being within a range of 20 degrees to 70 degrees with respect to an inner axis of the containment ring. Of course, other ranges greater or less the 20 degrees to 70 degrees with respect to an inner axis of the containment ring are considered to be within the scope of various embodiments of the present disclosure. In one further non-limiting embodiment, the ribs **54** are angled at 45 degrees with respect to the axis of rotation **20**. Of course, other angles greater or less than 45 degrees with respect to the axis of rotation **20** are considered to be within the scope of various embodiments of the present disclosure. This configuration of the inner portion or inner containment zone **52** is designed to absorb the kinetic energy of disc fragments **56** in the event of a tri-hub fracture event. For example, in the event the rotor **26** has separated into fragments illustrated in FIG. **8** the fragments will contact some of the ribs **54** and the ribs **54** will deflect radially outward due to their angular configuration and a plurality of openings **58** located between each of the ribs **54**. As such, the deflection of the ribs **54** and an openings **58** will create pockets during absorption of the kinetic energy of the fragments into the containment ring **34**. Note the description of a tri-hub fracture event is merely provided as an example and various embodiments of the present disclosure are not limited to this specific event. By providing a plurality of ribs **54** in the inner portion or inner containment zone **52**, the ribs **54** through their kinetic energy absorption reduces the impact force applied to the solid outer portion or outer containment zone **50**. As such, a thickness of the outer portion or outer containment zone **50** can be reduced while still providing the required containment. This reduction in thickness allows for a reduction in weight of the containment ring **34** which is desirable in the gas turbine engine **10**.

(20) As such, the outer portion or outer containment zone **50** is solid while the inner portion or inner containment zone **52** includes the ribs **54** and the outer portion or outer containment zone **50** of the containment ring **34** is considerably reduced in thickness, reducing the weight of the engine **10** while still containing the rotor fragments **56**.

(21) As mentioned above a plurality of openings **58** are located between each of the plurality of ribs **54**. FIGS. **6** and **7** are cross-sectional views of the containment ring **34** in a direction transverse or orthogonal with respect to the cross sectional views illustrated in FIGS. **4** and **5**. FIG. **6** being a cross-sectional view across of one of the ribs **54** and FIG. **7** is a cross-sectional across one of the openings **58**.

(22) As illustrated, the plurality of openings **58** do not extend all the way through the forward end

44 and aft end 46 of the containment ring 34. Nor do these openings 58 extend to the outer periphery 40. In other words, the openings 58 only extend from or open into the inner periphery 42. In addition, and in one embodiment, the plurality of openings 58 are scalloped shaped such that each opening 58 has an inner or bottom surface at its greatest depth from the inner periphery 42 at a mid portion or central portion 60 of the opening 58. In this embodiment, the depth of each opening 58 gradually decreases as it extends from the mid portion or central portion 60 towards the forward end 44 and aft end 46 of the containment ring 34. In other words and in one embodiment, each opening 58 has a curved bottom surface that extends from the mid portion or central portion 60 towards the forward end 44 and aft end 46 of the containment ring 34. Of course, other configurations may be employed for openings 58.

(23) In one non-limiting embodiment and shown in FIGS. 6 and 7, an axial length of the inner periphery 42 from the forward end 44 to the aft end 46 is greater than an axial length of the outer periphery 40 from the forward end 44 to the aft end 46 such that the main body portion 49 of the containment ring 34 has a trapezoidal shape in its cross-section or in other words the forward end 44 and the aft end 46 are angled surfaces that extend towards each other from the inner periphery 42. As such, the inner periphery 42 has a greater surface area than the outer periphery 40. Of course, other configurations of the cross-sectional shape of the containment ring 34 are considered to be within the scope of the present disclosure. Non-limiting examples include rectangular, square, any equivalents thereof, etc.

(24) Also shown is that the flange portion 48 of the containment ring 34 is secured to the main body portion 39 via a plurality of tabs 62. In one embodiment, the tabs 62 are separated from each other by a plurality of openings 64. In one embodiment, the tabs 62 are configured to be frangible when the containment ring 34 is impacted with disc fragments during a fracture event. Still further and in one non-limiting embodiment, the tabs 62 are integrally formed with the main body portion 39 of the containment ring 34 so that they (e.g., the main body portion 39) and the flange portion 48 form a single unitary structure.

(25) FIG. 8 is an illustration of containment of portions 56 of the rotor 36 by the containment ring 34 in accordance with the present disclosure.

(26) In accordance with the present disclosure, the containment ring 34 architecture with a ribbed inner portion and solid outer portion is provided in order to maximize flexibility and energy absorption of the containment ring 34 in particular during a tri-hub fracture event of the rotor 26.

(27) For example and in an uncontrolled overspeed event, rotating parts (turbine discs, compressor rotor, impeller, etc.) can fracture in three parts (tri-hub) and the high energy fragments are contained by the containment ring 34. See for example, FIG. 8. As such and in accordance with the present disclosure, these fragments are contained using the containment ring 34 of the present disclosure.

(28) When the fragment(s) 56 hit(s) the ring 34, the kinetic energy (KE) is absorbed by the potential energy of the ring (PE) in particular, the inner portion or inner containment zone 52 (e.g., ribs 54) which deform upon impact. As such, the inner portion or inner containment zone 52 with its ribs 54 and openings 58 causes the inner portion or inner containment zone 52 to be more flexible and plastically deform locally at an impact point which also maximizes the containment ring's 34 energy absorption. Thus, the inner portion or inner containment zone 52 acts as the energy-absorbing sacrificial zone whereas the outer portion or outer containment zone 50 contains the fragments and prevents them from being released outside the outer structural case 32.

(29) The term "about" is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, "about" can include a range of $\pm 8\%$ or 5% , or 2% of a given value.

(30) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates

otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(31) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A containment ring for a turbine casing assembly, comprising: a main body portion, the main body portion having an outer periphery and an inner periphery and a forward end and an aft end each extending between the outer periphery and the inner periphery, the main body portion having a center axis and also including an outer containment zone and an inner containment zone, the inner containment zone is integrally formed with the outer containment zone in order to provide a single unitary structure and the inner containment zone being radially inward from the outer containment zone, the inner containment zone extending radially outward from the inner periphery and having a plurality of ribs, each of the plurality of ribs extending continuously from the forward end and the aft end and each of the plurality of ribs being substantially parallel to the center axis, wherein the plurality of ribs are each separated by an opening extending into the inner periphery, the opening does not extend to the outer periphery or the forward end or the aft end and each of the plurality of ribs have opposing surfaces that are anularly offset in the same direction with respect to the inner periphery of the containment ring and to a plane extending radially from the center axis.
2. The containment ring as in claim 1, wherein the plurality of ribs are equally spaced from each other.
3. The containment ring as in claim 1, further comprising a flange portion secured to the main body portion.
4. The containment ring as in claim 3, wherein the flange portion is secured to the main body portion via a plurality of tabs.
5. The containment ring as in claim 1, wherein the main body portion is circular in shape.
6. The containment ring as in claim 1, wherein the opening extends from the inner periphery and has a scalloped shape.
7. The containment ring as in claim 1, wherein the opening has an inner surface that is furthest from the inner periphery at a central portion of the opening and a depth of the opening from the inner periphery gradually decreases as it extends from the central portion towards the forward end or the aft end.
8. The containment ring as in claim 7, wherein the opening has a curved bottom surface that extends from the central portion towards the forward end and the aft end of the main body portion.
9. The containment ring as in claim 1, wherein an axial length of the inner periphery from the forward end to the aft end is greater than an axial length of the outer periphery from the forward end to the aft end.
10. The containment ring as in claim 1, wherein the main body portion has a trapezoidal shape in its cross section.
11. The containment ring as in claim 1, wherein the forward end and the aft end are angled surfaces that extend towards each other from the inner periphery.

12. A turbine casing assembly, comprising: an outer structural case; and a containment ring, comprising: a main body portion, the main body portion having an outer periphery and an inner periphery and a forward end and an aft end each extending between the outer periphery and the inner periphery, the main body portion having a center axis and also including an outer containment zone and an inner containment zone, the inner containment zone is integrally formed with the outer containment zone in order to provide a single unitary structure and the inner containment zone being radially inward from the outer containment zone, the inner containment zone extending radially outward from the inner periphery and having a plurality of ribs, each of the plurality of ribs extending continuously from the forward end and the aft end and each of the plurality of ribs being substantially parallel to the center axis, wherein the plurality of ribs are each separated by an opening extending into the inner periphery, the opening does not extend to the outer periphery or the forward end or the aft end and each of the plurality of ribs have opposing surfaces that are angularly offset in the same direction with respect to the inner periphery of the containment ring and to a plane extending radially from the center axis.

13. A gas turbine engine, comprising: a compressor section; a combustor; a turbine section; and a turbine casing assembly, comprising: an outer structural case; and a containment ring, comprising: a main body portion, the main body portion having an outer periphery and an inner periphery and a forward end and an aft end each extending between the outer periphery and the inner periphery, the main body portion having a center axis and also including an outer containment zone and an inner containment zone, the inner containment zone is integrally formed with the outer containment zone in order to provide a single unitary structure and the inner containment zone is radially inward from the outer containment zone, the inner containment zone extending radially outward from the inner periphery and having a plurality of ribs, each of the plurality of ribs extending continuously from the forward end and the aft end and each of the plurality of ribs being substantially parallel to the center axis, wherein the plurality of ribs are each separated by an opening extending into the inner periphery, the opening does not extend to the outer periphery or the forward end or the aft end and each of the plurality of ribs have opposing surfaces that are angularly offset in the same direction with respect to the inner periphery of the containment ring and to a plane extending radially from the center axis.
